

EnLFADE: Ensemble Learning Based Fake Account Detection on Ethereum Blockchain

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Abstract:

Cryptocurrencies continue to captivate businesses and investors despite market fluctuations. The number of crypto users have risen rapidly in the last few years, and alarmingly, many appear to be unaware of the risks involved. These risks aren't confined to market hazards but include very sophisticated cybercrimes related to cryptocurrencies. As cryptocurrencies have become a breeding ground for a variety of cybercrimes, resulting in enormous financial losses, it hinders user adoption limiting the utility of the blockchain technology. It has become crucial to spot such scams and devise intelligent techniques to make this technology a safer place for investors. This study proposes a classification model to handle fake account problem over Ethereum blockchain, and its contribution is multi-faceted; firstly, available imbalanced Ethereum dataset has been balanced to enhance the accuracy of the classification model. Secondly, correlation-based feature selection technique has been applied to retain best discriminating features. Thirdly, an effective machine learning based model has been presented for the identification of fake accounts over the Ethereum system. A comparative study of ten machine learning techniques has been presented consisting of both individual and ensemble classifiers. Experimental results showed that ensemble classifiers appear to yield better performance measures over individual classifiers and among all, LightGbm-based classification model outperformed with 99.2 % accuracy.

Keywords: Ethereum Blockchain, Ensemble Learning, Machine Learning, Data Imbalance, Fake Account Detection

1. Introduction

Cryptocurrencies have seen a massive growth due to the evolution of blockchain technology and its economic ecosystem. Latest innovations in digital currencies have given rise to a whole new means of exchanging values that has the potential to eventually replace traditional notary services or payment processing organisations [1, 2]. Hundreds of cryptocurrency exchanges have emerged as essential trading platforms for the ecosystem, facilitating the trading of digital assets. As of the end of 2018, there were over 2,000 distinct cryptocurrencies, including Bitcoin, Ethereum, and Litecoin [1]. However, the current technical advancement of cryptocurrencies and its associated advantages have been tainted by a several unlawful affairs taking place on the network, including illicit accounts, money laundering, bribery and phishing etc. [3], resulting in massive financial losses [4].

1.1 Cryptocurrency and blockchain:

A cryptocurrency or a virtual currency is a digital asset designed to function as a medium of exchange that encrypts transactions. A cryptocurrency's transactions are given substance by a blockchain that accumulates and archives all trades between the two parties permanently. Each record is referred as a block that includes a digital hash pointer to the earlier block, a timestamp, and trade records [5]. A blockchain usually implements a cryptocurrency which can be swapped for other cryptocurrencies or fiat money through exchanges. For years, the data and information of people have long been held and performed by a centralised and trusted arbitrator, such as governments or businesses. Due to organisational changes, people have to confront issues like excessive fees, data silos, and instability.

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However, after the induction of blockchain technology, one doesn't need to rely on these old-fashioned third parties as the data can be effortlessly distributed around the world. The technology both makes the system reliable and reduces the charges levied by middlemen [6]. Thus, cryptocurrency is comparable to traditional currency, with the exception that it does not have a physical form and operates through encryption [46]. A quick rundown of the two most prominent cryptocurrencies are as follows:

Bitcoin: Created in 2009, Bitcoin is the first digital money to run in a multi user architecture, with no central authority to monitor transactions. It is the first cryptographic commodity that can be traded like cash. It has risen to become the world's most popular cryptocurrency in recent years, with an increasing number of businesses accepting it as payment, fostering a crypto-driven society [2, 7].

Ethereum: Introduced in 2014 by Vitalik Buterin, Ethereum is a publicly available blockchain platform that supports smart contracts. It facilitates multi user transactions by utilising Ether, a dedicated cryptocurrency [6]. Like Bitcoin, Ethereum is based on blockchain technology, but it goes one step further by allowing developers to execute programmes, such as smart contracts, that may host any type of decentralised application, often known as "dApps" [47]. Thus, Ethereum provided solution to the challenges faced in Bitcoin. In line with [49], buying ether money is significantly faster than buying bitcoin, around 14-15 seconds compared to bitcoin's near-uniform 10 minutes.

1.2 Frauds in cryptocurrencies

The invention of digital currency, specifically Bitcoin and Ethereum, was initially hailed as a game-changer; nevertheless, the popularity of these currencies has grown in tandem with a bad spirit of speculation, fraud, and hype throughout the past decade. Bitcoin investors were duped by a Ponzi scheme known as the Bitcoin Savings & Trust. In this scheme, early investors were given money by taking money from later investors. It became one of the most infamous fraud cases using a fictitious Initial Coin Offering (ICO) [7, 48]. Similar as Bitcoin, Ethereum has also been the victim of multiple cyberattacks in tandem with its growing popularity. It has also been subjected to several entities engaging in illicit behaviour on the network, including phishing, money laundering, fraud, and other criminal activities [3]. Despite the fact that the bitcoin technology is continually changing and trying to provide an error-free system for its users, the reputation of cryptocurrency has been tarnished as a result of the aforementioned behaviours. As mentioned in [3], an anomalous behaviour over the network was discovered after an account paid \$450,000, divided in 3 trades, to transfer almost 0.1 ETH in 2019. Such transactions raised questions about their legitimacy, further highlighting if they were genuine errors in which the account owner mistaken the transaction fee for transfer of Ether, or if they are part of any conspiracy in regard to money laundering [3]. The proliferation of the aforementioned Ethereum scams will not only stifle user adoption, but will also limit block chain technology's utility. As a result, it is critical to identify such scams and develop effective strategies to make technology a safer environment for investors.

Recent enhancement in Machine Learning (ML) has attracted researchers worldwide to use data and algorithm and train machines to learn in the same way that humans do, with the goal of constantly improving accuracy. With the rise in Ethereum transactions, ML has become a major tool for handling classification difficulties. Such algorithms adaptively improve their performance as the number of transactions available for learning increases. ML techniques consists of both individual and ensemble learning based classification. The ensemble learning strategy combines multiple base models to create a single optimal predictive model. When compared to individual models, ensemble learning strategies have been shown to perform better and have higher predictive accuracy. Current study has demonstrated the applicability of ML techniques focusing on five different individual classifiers including Naïve Bayes [9], Decision Trees [10], Support Vector Machine (SVM) [11], K Nearest Neighbour (KNN) [12], Logistic Regression [13] and five different bagging and boosting Ensemble Classifiers including Catboost [14], Adaboost [15], Xgboost [16], LightGbm [17] and Random Forest [18]. However, for an ensemble ML classifier to function effectively, experimental dataset needs to be balanced as in many cases class label attribute is not equally distributed which further contributes to the creation of biased models. In this work, a popular technique known as Synthetic Minority

Oversampling Technique (SMOTE) [8, 32] has been used for handling imbalanced dataset to improve accuracy. As per our knowledge, this study is the first systematic work of the application of ensemble techniques for the detection of fake accounts over Ethereum blockchain. The work can be utilised to detect fake accounts by investigative and enforcement authorities, developers, researchers, news organisations, and private companies, and remains very helpful in performing filtered activities like combating terror financing and money laundering [4]. Major contributions of this paper includes:

- To design an ensemble ML based classification model for classifying fake accounts over the Ethereum network.
- Comparative analysis of various ensemble classifiers' performance over individual classifiers.
- To perform exploratory data analysis and handling imbalanced dataset using SMOTE.
- Feature selection using Pearson Correlation Coefficient for enhancing accuracy.

The remaining paper has been laid out as follows: Section 2 gives an assessment of the related works. Section 3 highlights ML ensemble approaches followed by section 4 which introduces the proposed ML based classification model and depiction of the dataset used in the reported study, pre-processing steps and feature engineering. Section 5 highlights the experimental and evaluated outcomes. Finally, Section 6 discusses the conclusion with future directions of work.

2. Related Works

This section shows some relevant study in which researchers proposed models for detecting cryptocurrency fraud. Researchers have been attempting to create computationally intelligent methods to detect frauds in the crypto space. Such frauds are growing rapidly with many crypto investors being duped either by a Ponzi scheme or the money invested in crypto is illegally transferred to a fraudulent account by online criminals. Pengcheng Xia et al. [1] were the first to identify and classify cryptocurrency exchange scams in 2019. They gathered existing reports and used typosquatting creation tools to identify approximately 1,500 and 300 scam domains and bogus apps, respectively. They then looked into their connections and discovered 94 fraudulent domain groups and 30 false app groups. Authors revealed that such scams resulted in a financial loss of \$520K and also discovered 183 blockchain addresses linked to the attackers. Their findings are critical for blockchain stakeholders, and they illustrate the urgency of identifying and preventing blockchain scams. In [19], Hao Hua Sun Yin et al. developed and validated a unique method for deanonymizing the Bitcoin network and recognized numerous unlawful activities using supervised learning during 2019. In this study, the dataset used includes nearly 395 million transactions among 12 categories. It also includes 957 unique cluster-created classifiers that distinguish between illegal activities, including scam stolen bitcoins, darknet market among others. The authors used seven Machine Learning algorithms viz. Decision Tree, Random Forest, Adaboost, Extra Trees, KNN, Bagging, and Gradient Boosting with Gradient Boosting algorithm outperforming the others with a mean cross validation accuracy and F1 score of 80.42 percent and 79.64 percent, respectively.

Further, T phan et al. [20] emphasized on anomaly detection over bitcoin network. The most suspect users and transactions were identified by the researchers in this investigation. Their dataset included over 6 million unidentified individuals, over 37 million transactions, and nearly 30 identified bitcoin criminals. They have created network of two graphs based on Bitcoin transaction, and employed various algorithms to discover anomalies namely - K-means clustering, Mahalanobis distance, and unsupervised Support Vector Machine (SVM). Despite their low agreement metrics, they were able to detect two known occurrences of burglary and one known case of loss out of a total of 30. Likewise, Sayadi et al. [21] proposed a study that employed two ML methods to identify bitcoin transaction fraud. To begin with, a one class SVM was identified to discover numerous irregularities in bitcoin electronic transactions and the accuracy was reported as 90%. Later, the authors utilised the k-means algorithm to find similar abnormalities.

It has been observed that Ethereum network is growing up nowadays leading to frauds similar as Bitcoin. Rahmez Fawaz Ibrahim et al. [22] in 2021, suggested a fraud detection model based on Decision Tree (J48), Random Forest, and K Nearest Neighbours (KNN) approaches for detecting bogus accounts over the Ethereum blockchain. The authors used a supervised dataset that included 7651 regular transactions and 159 fraudulent transactions with 42 attributes. They were able to enhance time measurements and F measure using Random Forest. In [3], Steven Farrugia et al. used datasets obtained from two sources: Etherscamb and a local Geth client, totalling 4681 accounts, with 2179 fraudulent and 2502 normal accounts. They proposed a gradient boosting based classifier Xgboost for the identification of bogus accounts. Authors claimed to attain an average accuracy of 96.4% and Area Under Curve (AUC) as 0.994, after applying 10-fold cross validation. Authors revealed that three features namely time difference between first and last minutes, total Ether balance and min value received outperform other features used in this study.

In Ethereum space, phishing scams are rapidly appearing that include creation of a fraudulent certified website and tracking accounts while logging, so as to steal sensitive data including passwords. In [23], authors proposed a systematic approach for detecting phishing accounts using Ethereum blockchain transactions. Transactions on the Ethereum network and the designated phishing addresses were collected by utilising the parity client and crawling etherscan.io. Authors developed a transaction graph utilising extracted data and suggested an attribute extraction technique. Then, they proposed a LightGbm-based Dual-sampling Ensemble model to detect phishing suspects based on the retrieved features. Yuan Huang et al. [4] developed a 3-phase framework by mining transaction records to detect Ethereum phishing frauds during 2020. Firstly, they built the Ethereum transaction network after obtaining the labelled phishing accounts and their transaction logs. Then, latent properties of accounts were extracted using node2vec for phishing classification. Finally, authors employed the one class SVM to determine whether the account is phishing. Their model attained an F score of 0.846. In [24], Alam Sarma et al. designed a framework to identify phishing scams using Random Forest (RF) and Decision Tree (DT). Their model was fed with standard datasets of phishing assaults obtained from kaggle.com. Further PCA was used to examine the features, and finally, they reported accuracy of 97% and 91% using RF and DT respectively.

Apart from phishing, various fraudulent behaviour has also been detected on Ethereum network. In [25], a study published during 2019 looked at etherscamb.info and etherscan.io databases and reported nearly 420 and 53 fraudulent wallets. Authors built a transaction network based on transactional data, evaluated it using graph traversal technique, and clustered them by applying ML techniques viz. K-means clustering, Support Vector Machine, and Random Forest. The Random Forest class model was reported as the most effective in their experiments.

From the above literature survey, it has been observed that most researchers have applied their learning on imbalanced datasets, resulting in bias model formation. In imbalanced dataset, standard classifiers are generally skewed by huge classes and the small ones are ignored. Further, it has been found that majority of the research work uses individual classifiers [20, 21, 22] to handle this problem. However, very few research efforts are based on ensemble techniques [3, 19]. The aim of the current work is to balance the dataset and exploit the benefits of ensemble learning techniques for achieving high precision and accuracy.

3. Methodology

Ensemble Machine Learning is a broad meta method that tries to improve predictive performance by merging the projections from various diverse models. For a number of classification and regression issues, ensemble approaches have proven to be effective. By combining several weak classifiers, the errors of a single base classifier will be remunerated by other strong base learners, leading to ensemble's overall prediction performance superior than that of a single base learner [27]. Such classifiers provide superior accuracy with lower errors, greater consistency by dodging overfitting, and decreases bias and variance errors. Bagging and boosting are the most popular ensemble ML techniques as they exhibit powerful experimental results and theoretical performance guarantees [28].

Bagging: A bootstrap aggregation technique developed by Breiman in 1996 initiates by voting various classifiers generated by different bootstrap samples. As shown in Figure 1, it learns several homogeneous weak classifiers with high variance (independently from each other) and combines their predicted result using averaging process in regression and majority voting in classification. The number of classifiers used in the ensemble has a direct relationship with the drop in this variance aspect [28]. Thus making it suitable for classification problems. The major advantages of bagging are as follows:

- To increase stability and accuracy of the bagged result.
- To reduce variance of the ensemble classifier.

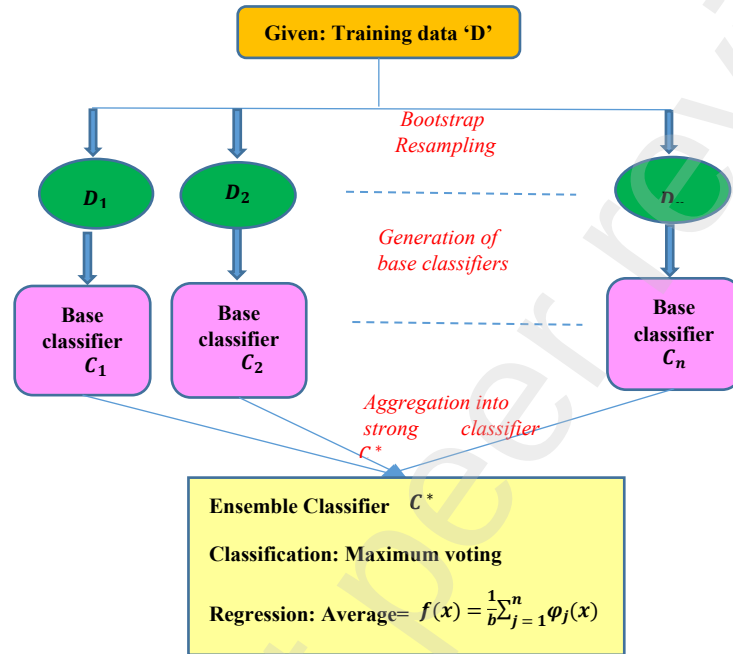


Figure 1: Bagging

Boosting: Boosting creates an ensemble of classifiers through adapting the training set's distribution based on the accuracy of previously constructed classifiers. The probability distribution is adjusted for samples that were mistakenly forecasted by prior classifiers can be chosen more often than those that were correctly forecasted. Working steps of Boosting can be visualized using Figure 2. Boosting aims to create new classifiers that can better categorise rigid samples for earlier ensemble members [28]. It has two main advantages:

- It has significant relevance for small sample sizes and high-dimensional samples, compared to other classification techniques.
- It can choose features while categorising and is quick, easy to programme, highly adaptable, and accurate [30].

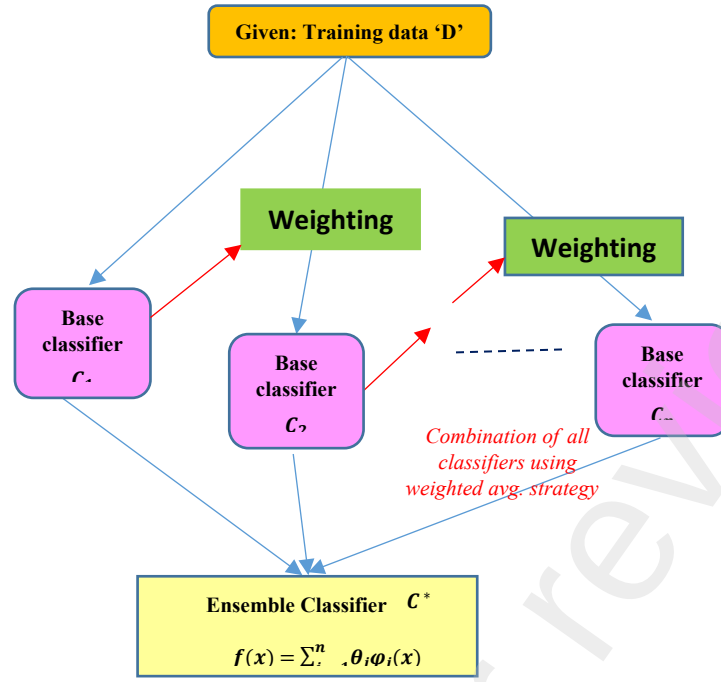


Figure 2: Boosting

Following sub-sections highlight the working details of some Bagging and Boosting algorithms used in this study.

3.1 Random Forest:

Random Forest [18], a most significant and robust Bagging method that can be seen as ensemble of classifiers made up of decision trees built from two different randomization sources. Each decision tree is proficient on a random sample of data, with replacement from original data of the equivalent size as the training set. Attribute sampling is a second source of randomization used in random forest [34]. Thus, bagging in conjunction with random attribute selection can be used to create random forests. It improves classification accuracy by using maximum voting and regulates the model to avoid overfitting [24]. In line with [33], the main steps of the algorithm are as follows:

1. Suppose $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$ denotes training data. A bootstrapping collection D_j of dimension N from D (with replacement) has been picked where $j = 1$ to J . With D_j as the training data, binary recursive partitioning has been used to adjust a tree.
 - a. Start by assembling all the data in a particular distinct node.
 - b. Till the terminating condition is achieved, repeat the following steps for each undivided node:
 - i. m predictors were randomly selected from the available p predictors.
 - ii. The efficient binary split has been chosen from all the m predictors
 - iii. Using the best suited split (from step ii) the node has been divided into two daughter nodes
2. At new point x , a prediction has been made for:

- Regression : $f(x) = \frac{1}{J} \sum_{j=1}^J \hat{h}_j(x)$ (1)

- Classification : $f(x) = \arg \max_y \sum_{j=1}^J I(\hat{h}_j(x) = y)$ (2)

where $\hat{h}_j(x)$ implies the prediction of x (response variable) using the j^{th} tree.

3.2 Adaboost

Freund and Schapire invented a boosting method known as Adaboost in 1996 [15], which creates a powerful learning model by combining multiple weak learning models. In each iteration, misclassified data points are identified and their weights are increased, while the weights of correct data points are

decreased, so that the following classifier can focus more on getting them correctly. The instruction continues in such a way that the weak learner's exit becomes the other learner's entry. The results are then merged, and the final decision limits are established [32, 50]. In this method, instances are pulled into successive data sets from an iteratively updated training data sample distribution, and the classifiers are integrated using weighted majority voting [31]. In line with [26], the major steps of the Adaboost algorithm are as follows :-

1. Suppose $X = \{(x_1, y_1), \dots, (x_m, y_m)\}$ be the sample set, where $x_i \in X$ (domain space) and $y_i \in Y$ where $Y = \{-1, +1\}$. Initialize,

$$D_t(i) = \frac{1}{m}, \text{ for } t = 1, \dots, T \quad (3)$$

Where $D_t(i)$ is a distribution that represents each instance's weights and train the weak classifier using it.

2. Training of weak classifier returns a weak hypothesis, $h_t: X \rightarrow \{-1, +1\}$ and the value of h_t should be set to minimise the weighted error:

$$\varepsilon_t = \sum_{i: h_t(x_i) \neq y_i} D_t(i) \quad (4)$$

3. Adaboost selects a parameter, α_t , to measure the relevance of the weak hypothesis and is defined as:

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1 - \varepsilon_t}{\varepsilon_t}\right) \quad (5)$$

4. Well-classified examples are given lower weights than those that are misclassified. Adaboost appraises the distribution for each x_i with label y_i , by taking into account α_t and h_t . $D_t = 1$ since Z_t is the chosen normalising factor.

$$D_{t+1}(i) = \frac{D_t(i)}{Z_t} \times \begin{cases} e^{-\alpha_t} & \text{if } h_t(x_i) = y_i \\ e^{\alpha_t} & \text{if } h_t(x_i) \neq y_i \end{cases} = \frac{D_t(i)(e^{-\alpha_t y_i h_t(x_i)})}{Z_t} \quad (6)$$

5. Adaboost returns a final hypothesis when all T iterations have been completed:

$$H(x) = \text{sign}\left(\sum_{t=1}^T \alpha_t h_t(x)\right) \quad (7)$$

3.3 Xgboost

Xgboost [16] is an efficient, versatile, and portable parallel decision tree-based ensemble ML approach [27, 32]. It applies gradient descent on decision trees in order to develop a number of models, that are sequentially integrated while the earlier models are rectified to obtain the ultimate suitable model [31]. Xgboost overcomes one of the most important shortcomings of gradient boosting i.e. the potential losses that can be encountered when branching the tree. It examines how features are distributed among all the data points in a leaf and uses this knowledge to branch out the data. Based on the strengthening method of the Xgboost algorithm, the basic idea is to build a weak model, make inferences regarding various feature importance and parameters, and then use these results to create a new, stronger model, attempting to reduce the existing error by exploiting the previous model's misclassification error [35, 43]. In order to prevent data over-fitting, Xgboost uses a more systematised model formalisation, which improves efficiency when compared to previous gradient boosting methods. To do so, there is a need to study h_i functions, each of which contains tree structure and leaf scores. In line with [22, 36], following are the major steps of the Xgboost algorithm :-

1. Given a dataset S having m -samples and n -features, $S = \{(X_j, y_j)\} (|S| = m, X_j \in \mathbb{R}^n, y_j \in \mathbb{R})$, a tree ensemble model employs L additive functions to forecast the result as presented in the equation (8).

$$y_j' = \varphi(X_j) = \sum_{l=1}^L h_l(X_j), h_l \in H \quad (8)$$

where $H = \{h(X) = w_q(X)\} (q: \mathbb{R}^n \rightarrow T, w \in \mathbb{R}^n)$ is the space of regression trees, q signifies each tree's structure that maps a sample to its corresponding leaf index, and T represents total leaves in the tree. Each h_l relates to independent structure of tree q and leaf weights w .

2. To understand the collection of functions used in the model, the regularised objective is minimised as follows in the equation (9).

$$L(\varphi) = \sum_j l(y_j, y_j') + \sum_l \Omega(h_l), \Omega(h) = \gamma T + \frac{1}{2} \lambda ||w||^2 \quad (9)$$

where l denotes differentiable convex loss function which calculates difference between the target y_j and predicted y_j' , Ω penalizes complexity of the model in order to elude over-fitting.

3. Xgboost also employs a 2nd order Taylor series goal function. If the loss or error function is the mean square error, then the objective function is stated using equation (10).

$$L(\varphi) = \sum_{j=1}^n [p_j w_{q(x_j)} + \frac{1}{2} (q_j w_{q(x_j)}^2)] + \gamma T + \frac{1}{2} \lambda \sum_{l=1}^L w_l^2 \quad (10)$$

where, p_j and q_j describes the 1st and 2nd derivatives of loss or error function, respectively, while $q(x_j)$ is a function which maps data points to the associated leaves.

4. Because the DT instances correspond to leaf nodes, the final loss value is the addition of loss values. Hence, objective function can be shown as:

$$L(\varphi) = \sum_{t=1}^T [P_t w_t + \frac{1}{2} (Q_t + \lambda) w_t^2] + \gamma T \quad (11)$$

where $P_t = \sum_{j \in I_t} p_j$, $Q_t = \sum_{j \in I_t} q_j$, and I_t indicates all samples in leaf node t .

5. To summarise, objective function optimization is reduced to challenge of obtaining the optimum value of a quadratic function. Due to the addition of the regularisation term, Xgboost is more resistant to overfitting.

3.4 LightGbm

Microsoft's LightGbm [17] is an open source gradient boosting decision tree method that implements a boosting technique in an efficient and scalable manner. Modern network connectivity is combined with the parallel voting decision tree method to maximise parallel learning. This method uses a histogram-based technique to speed up training and consume less memory. To locate the highest splitter gain of the leaf, LightGbm employs a leaf-by-leaf approach [37]. It employs two unique techniques: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to handle vast data samples and features respectively. GOSS preserves all instances having higher gradients and random samples with lesser gradients. The EFB method aims at combining numerous selective features into a smaller number of dense characteristics, reducing the amount of time spent calculating zero feature values [34]. Large volumes of data can be processed quickly with the LightGbm algorithm. In line with [17, 42], the major steps of the algorithm are given as follows :-

1. Suppose training data $S = \{ (x_1, y_1), \dots, (x_N, y_N) \}$, $x_i \subseteq \mathbb{R}$, $y_i \in Y = \{-1, +1\}$, loss function be $L(y, x)$ and number of iterations be M . Let the sampling ratio of large gradient data and sampling ratio of small gradient data be a and b respectively.

2. Merge mutually exclusive features by using EFB method. For $m=1$ to M , calculate absolute values of gradients $r_i = \left| \frac{\partial L(y_i, \theta(x_i))}{\partial \theta(x_i)} \right|_{\partial \theta(x) = \theta_{m-1}(x)}$, $i = \{1, \dots, N\}$ (12)

3. Resampling dataset by using GOSS method:
top $N = a \times \text{len}(S)$ and rand $N = b \times \text{len}(S)$,
sorted = GetSortedIndices(abs(r)),
 $E = \text{sorted}[1 : \text{top } N]$, $F = \text{RandomPick}(\text{sorted}[\text{top } N : \text{len}(S)], \text{rand } N)$
 $S' = E + F$

4. Compute information gains: -

$$V_j(d) = \frac{1}{n} \left(\frac{(\sum_{x_i \in A_l} r_i + \frac{1-a}{b} \sum_{x_i \in B_l} r_i)^2}{N_l^j(d)} + \frac{(\sum_{x_i \in A_r} r_i + \frac{1-a}{b} \sum_{x_i \in B_r} r_i)^2}{N_r^j(d)} \right)^{nt} \quad (13)$$

5. Obtain another decision tree $\theta_m'(X)$ on set S' . Then update $\theta_m(X) = \theta_{m-1}(X) + \theta_m'(X)$, for $m=1$ to M . Hence, $\theta_M'(X) = \theta_M(X)$

3.5 Catboost

It has been observed that, CatBoost [14] may be more efficient when applied to dataset that comprises a huge number of categorical values having large cardinality in addition to numerical attributes. The name, CatBoost, is a mixture of the terms "Category" and "Boosting", which is a Yandex-developed open source ML technique that effectively manages categorical attributes and outdoes previous publicly available gradient boosting methods in terms of performance [50]. As given in [39] learning and scoring methods are implemented on the GPU and CPU respectively, making the library significantly faster than previous gradient boosting libraries. The ability to operate with both numerical, category, and text data, as well as GPU support and visualisation choices, distinguishes this approach from others. It has numerous benefits:

- A novel approach is built in to treat category attributes as numerical characteristics automatically.
- It employs a blend of category attributes that capitalise on feature linkages, considerably enhancing feature dimensions.
- To prevent overfitting and increase the algorithm's accuracy and generalizability, a balanced tree model has been used [38].

4. Proposed Model

This paper proposed an ensemble learning based classification method known as **EnLFADE** for the detection of fraudulent transactions over the Ethereum blockchain. The various components of the model have been presented in Figure 3. It consists of four different functional components – data collection, data pre-processing, model building & training and model validation & evaluation. These components have been presented in detail in the following subsections.

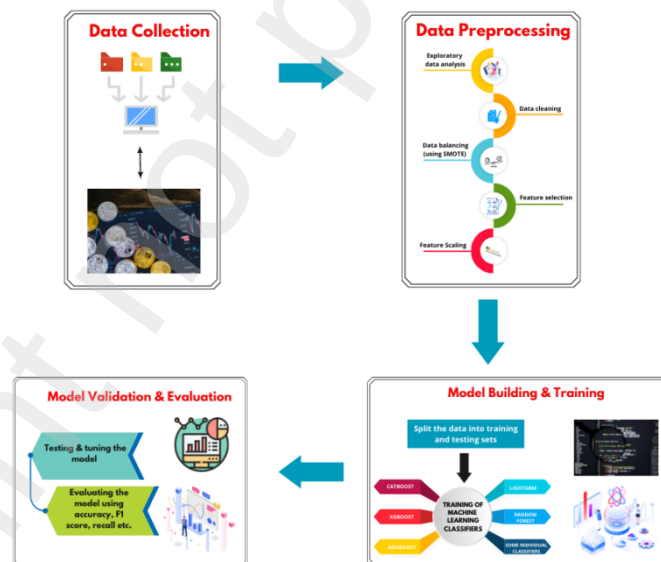


Figure 3: EnLFADE- Proposed Ethereum fake account detection model.

4.1. Dataset

The supervised Ethereum dataset has been collected from [3, 22, 44], containing 9841 transactions: 7662 normal transactions (77.9%) and 2179 fraudulent ones (22.1%). It consists of 51 attributes and is a two-class problem with class label 'genuine account' and 'fake account' respectively. The genuine account class describes an account with legal and normal transactions, whereas, illicit account class specifies an account with fraudulent and fake transactions. The dataset displays rows of recognised illegal and legal transactions built over Ethereum network. Table 1 gives the details of the dataset used in this work.

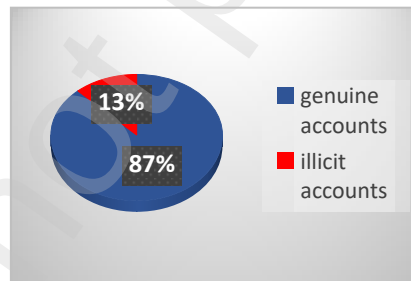
Table 1: Details of Ethereum dataset

Dataset	No. of samples	Output classes	Input attribute	Output attribute	Least important attributes	Total no. of attributes
Ethereum dataset	9841	2	50	1	17	33

Extracted dataset contains some duplicate features and a few having variance 0. Hence, 17 out of 51 unwanted attributes have been dropped as they will not contribute significantly in the performance of the model. Remaining 34 attributes are retained for further analysis. Details of the same has been discussed in the section 4.3.

4.2 Data pre-processing

On performing Exploratory Data Analysis (EDA), it is observed that data requires cleaning, balancing and scaling. After dropping null and duplicate values, 8739 data instances were left for further experimentation. As shown in Figure 4, the target class label is not equally distributed leading to imbalanced dataset. In case of imbalanced dataset, the model is trained mainly on the majority class and it becomes biased towards the majority class prediction. In this study, SMOTE [8, 32] has been used for managing the problem of class imbalance. This method produces new data samples of the illicit account class by emulating prevailing minority samples (i.e. illicit account) and making minor changes internally. The usage of SMOTE found that the number of samples for both the classes were same. Moving forward, in order to handle highly varying magnitudes, it is important to perform feature scaling on the data. Without scaling features, the algorithm may be biased towards the features which have higher magnitudes. In this study, standard scaler method has been used. It's a technique for regulating the data's autonomous features within a given range.

**Figure 4:** Imbalanced Ethereum dataset

4.3 Feature selection

Feature selection not only removes the unwanted features but also helps in finding the most effective discriminator in the classification problem. It further helps in improving performances, achieving higher accuracies and decreasing models' logical errors. In this paper, Pearson Correlation Coefficient [35] has been applied to choose the most crucial features. It is a statistical number that highlights the dependency between two features and is measured using the following formula:

$$\rho_{x,y} = \frac{\text{cov}(x,y)}{\sigma_x \cdot \sigma_y} = \frac{E(xy) - E(x)E(y)}{\sqrt{E(x^2) - E^2(x)} \cdot \sqrt{E(y^2) - E^2(y)}} \quad (14)$$

where, $\text{cov}(x, y)$ implies covariance. Further, $\sigma_x \cdot \sigma_y$ depicts the product of vector standard deviation and $\rho_{x,y} \in \{-1, +1\}$. Stronger positive correlation and stronger negative correlation is shown by value close to +1 and -1 respectively. Figure 5 displays the heatmap of the correlation matrix of 33 features used in this study.

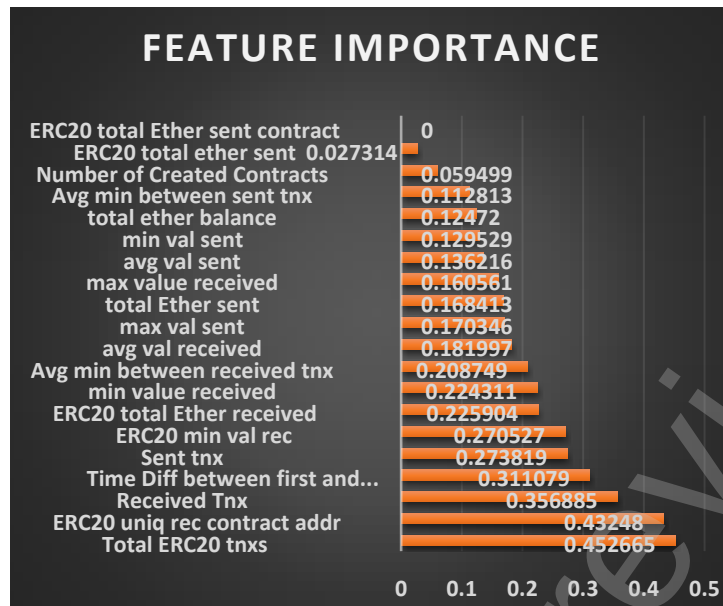


Figure 6: The ranking of top 20 features based on Mutual Information

4.4 Model learning and classification

In the initial phase of the experiment, classification of data has been achieved using five individual classifiers. Further, to achieve better classification accuracy, five different bagging and boosting ensemble classifiers are applied. List of the same are shown in the Figure 7.

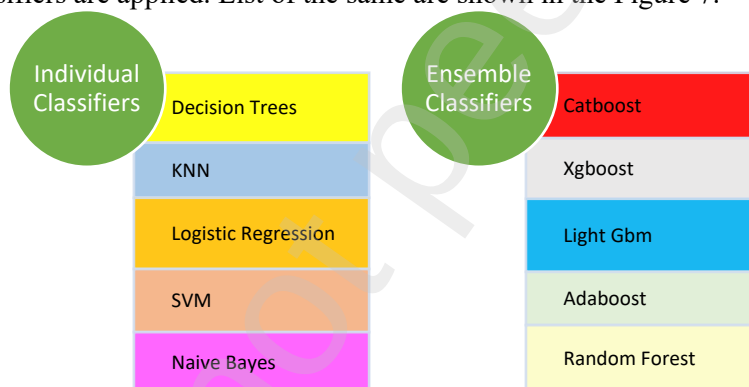


Figure 7: Individual and ensemble classifiers

4.5 Hyper parameter tuning

Top-level parameters useful for configuring a Machine Learning based model are known as hyper parameters. An efficient strategy for selecting the set of optimal hyper parameter values for a learning algorithm in order to have the best modelling output is known as optimization or hyper parameter tuning. In line with [40, 41, 34], description of various hyper parameters that have been tuned in the proposed work are given as follows:

- **base_estimator:** It is the base learner from which the boosted ensemble is built.
- **n_estimator:** The maximum number of trees to train sequentially
- **random_state:** It is the integral value which controls the random seed given at each base estimator at each iteration.
- **max_depth:** It is defined as maximum depth of the tree. It limits the complexity and depth of the trees.
- **criterion:** It checks the quality of the split, e.g. Gini, entropy etc.
- **min_sample_split:** The minimum number of samples required to split an internal node
- **max_features:** The number of features required for the best split
- **min_child_samples:** Minimum samples required in the child leaf.

- **learning_rate:** It is considered as crucial parameter that controls the rate or speed at which model learns and is also defined as the weight applied to each classifier at each boosting iteration. Also known as shrinkage.
- **num_leaves:** It is the maximum number of leaves per tree.
- **subsample:** The subsampling rate for the size of the random samples.

4.6 Performance evaluation

This study contains several evaluation metrics including confusion matrix, accuracy, recall, specificity, precision, F1 score and log loss. These precise criteria are compared to reveal the most effective classifier used in this work.

- **Confusion matrix:** A 2 x 2 matrix used to determine the link between the model's expected and actual value.

Actual Label	Predicted label	
	Negative or 0 or genuine	Positive or 1 or fake
	Negative or 0 or genuine	Positive or 1 or fake
Negative or 0 or genuine	True Negative (TN) (refers to the case in which the classifier predicted that the account is genuine, and actually it is genuine)	False Positive (FP) TYPE:1 ERROR (refers to the case in which the classifier predicted the account as fake, but actually it is not fake)
Positive or 1 or fake	False Negative (FN) TYPE:2 ERROR (refers to the case in which the classifier predicted that the account is genuine, but actually it is fake.)	True Positive (TP) (refers to the case in which the classifier predicted that the account is fake, and actually it is fake.)

- **Accuracy:** It is defined as occurrence of correct predictions w.r.t. all the available examples being tested. The accuracy metric calculates the percentage of cases that are successfully classified.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

(16)

- **Sensitivity (Recall):** It is described as the proportion of genuine positives (fake accounts) to all actual positives.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

(17)

- **Specificity:** It is the fraction of negative class (i.e. genuine accounts) got correctly classified.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

(18)

- **Precision:** It determines positive class measurement (i.e. illicit accounts) that are being truly acknowledged by definite classifier besides demonstrating the definite propriety.

$$\text{Precision} = \frac{TP}{TP + FP}$$

(19)

- **F1 score:** The F_β coefficient helps in assessing the predictive effectiveness of the model based on the outcomes from both Recall and Precision. It is defined as:

$$F_\beta = \frac{(1 + \beta^2)xy}{(\beta^2x) + y}, \text{ where } x \text{ is precision, and } y \text{ is recall.} \quad (20)$$

$$\therefore F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} = \frac{2xy}{x + y} \quad (21)$$

- **Log Loss:**

Log-loss, or logarithmic loss, tells about the minute information of ML classifier. As stated in [45], "Log-loss is a 'soft' measurement of accuracy that incorporates this idea of probabilistic confidence". Hence, the binary classifier's Log-loss is:

$$\text{Log-loss} = -\frac{1}{N} \sum_{i=1}^N y_i \log b_i + (1 - l_i) \log(1 - b_i) \quad (22)$$

where b_i depicts the probability that the i^{th} instance belongs to class 1, as judged by the classifier. $l_i \in \{0,1\}$ which depicts the true label. Log-loss is the cross entropy between the distribution of the true labels and the predictions. Thus, lesser the value of Log loss or cross entropy, more accurate the classifier is.

5. Experimental and Evaluation Results

The findings of the proposed machine learning-based methodology in effectively detecting illegitimate accounts are summarised in this section. During the experimental phase, total six major observations have been evaluated as shown in the following subsections:

Observation 1: As previously stated, the dataset was initially imbalanced, and SMOTE has been used to balance it in order to improve accuracy and reduce model bias. Table 2 compares different ML classifiers trained on a balanced dataset versus an imbalanced dataset, both having 33 features. The accuracy metric is ineffective in the imbalanced data situation since the classifier can still achieve higher accuracy if it predicts all instances as negative. As a result, F1 score has been computed instead of accuracy in this table because it provides a more comprehensive picture of the classifiers' performance.

Table 2: Comparison of F1 scores on imbalanced and balanced dataset

	Classifiers	Class label 0(normal) 1(fraud)	Imbalanced dataset (33 features)	Balanced dataset (33 features)		Classifiers	Class label 0(normal) 1(fraud)	Imbalanced dataset (33 features)	Balanced dataset (33 features)
1	Catboost	0	.99	.99	6	Decision trees	0	.99	.97
		1	.95	.99			1	.88	.97
2	Xgboost	0	.99	.99	7	KNN	0	.94	.96
		1	.94	.99			1	.57	.96
3	Light Gbm	0	.99	.99	8	Logistic Regression	0	.94	.68
		1	.96	.99			1	.03	.78
4	Adaboost	0	.99	.99	9	Svm	0	.94	.83
		1	.94	.99			1	0	.87
5	Random forest	0	.99	.99	10	Naïve bayes	0	.94	.31
		1	.94	.99			1	0	.71

From Table 2, it can be concluded that the balanced dataset provides better and unbiased results. So the balanced data was randomly divided into two parts, 80% being used for model training and 20% for model testing.

Observation 2: Table 3 and Figure 8 and 9 illustrate the comparison of performances for all classification algorithms applied on the balanced Ethereum dataset with 20 selective features containing 12211 samples from the perspective of Accuracy, Precision, Recall, F1 score, Specificity and Log loss.

Table 3: Performance comparison of various classifiers on Ethereum dataset

		Classifiers	Accuracy (in %)	Precision	Recall (sensivity)	F1 score	Specificity	Log loss
1	Ensemble Classifiers	Catboost	99.12	.99	.99	.99	.99	.31
2		Xgboost	99.18	.99	.99	.99	.99	.28
3		LightGbm	99.2	.99	.99	.99	.98	.25
4		Adaboost	98.8	.98	.99	.99	.98	.41
5		Random Forest	98.9	.98	.99	.99	.98	.4
6	Individual Classifiers	Decision Trees	97.1	.97	.97	.97	.97	.9
7		Knn	95.1	.93	.97	.95	.93	1.6
8		Logistic Regression	73.6	.69	.88	.77	.57	9.1

9		Svm	82.2	.77	.92	.84	.71	6.1
10		Naïve Bayes	59.8	.56	.9	.71	.19	13.8

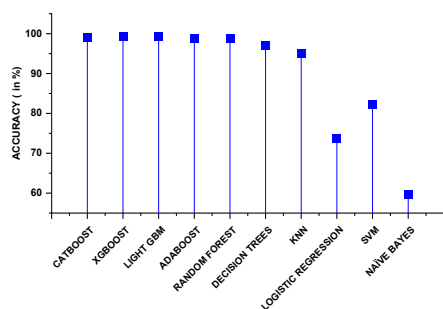


Figure 8: Accuracy (in %) of different classifiers

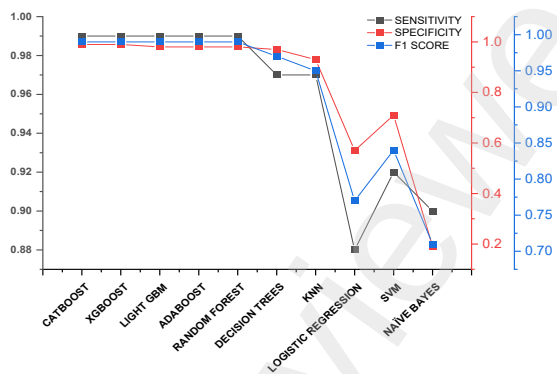


Figure 9: Sensitivity, Specificity, F1 score of different classifiers

From Table 3, Figure 8 and 9, it can be observed that ensemble classifiers - Catboost, Adaboost, Xgboost, LightGbm and Random Forest are performing better than the individual classifiers. The parameter setting of ensemble classifiers is illustrated in Table 4.

Table 4: Parameter setting of Ensemble Classifiers

Adaboost			LightGbm		
Parameters	Default Values	Optimal Values	Parameters	Default Values	Optimal Values
Algorithm	'SAMME.R'	'SAMME.R'	Boosting type	'gbdt'	'gbdt'
Base Estimator	none	Random Forest Classifier	Number of leaves	31	31
Learning Rate	1	0.01	Max depth	-1	-1
N Estimators	50	100	Learning rate	0.1	0.2
Random State	none	96	N estimators	100	100
			Min child samples	30	20

Catboost			Random Forest		
Parameters	Default Values	Optimal Values	Parameters	Default Values	Optimal Values
Learning rate	0.03	0.2	Criterion	'gini'	'gini'
Max depth	6	8	Min samples split	2	3
Leaf estimation iterations	none	10	Max features	'auto'	'auto'
N estimators	500	300	Random state	None	10

Xgboost		
Parameters	Default Values	Optimal Values
Learning rate	0.1	0.1
Max features	'auto'	'auto'
Max depth	3	8
N estimators	100	300

Observation 3: Figure 10 illustrates confusion matrices of different 10 ML classifiers.



Figure 10: Confusion matrices for (a) LightGbm (b) Catboost (c) Adaboost (d) Random Forest (e) Xgboost (f) KNN (g) Decision Trees (h) Naïve Bayes (i) SVM (j) Logistic Regression

Observation 4: Table 5 and Figure 11 show the FN, FP, TP and TN score of different classifiers obtained from confusion matrices shown in Figure 10. The aim of an efficient model is to maximize TP, TN and minimize FN, FP. Also, in case of illicit account detection problem, if a fraudulent transaction goes undetected then it will incur a huge loss. Thus, focus should be to minimize FN.

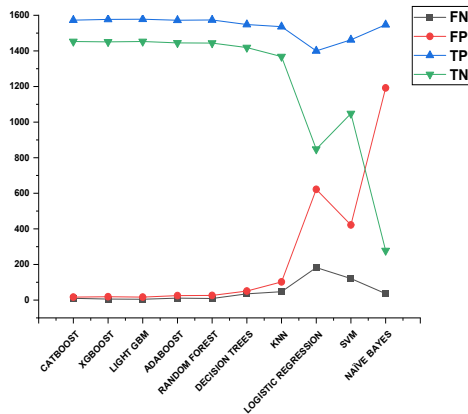


Figure 11: FN, FP, TP and TN score of different classifiers.

Table 5: FN, FP, TP and TN score of different classifiers

	FN	FP	TP	TN
Catboost	10	17	1573	1453
Xgboost	6	19	1577	1451
LightGbm	5	17	1578	1453
Adaboost	11	25	1572	1445
Random Forest	9	26	1574	1444
Decision Trees	35	51	1548	1419
Knn	47	102	1536	1368
Logistic Regression	183	622	1400	848
Svm	121	422	1462	1048
Naïve Bayes	36	1192	1547	278

5.5 Observation 5: Figure 12 showcases the ROC curve and AUC score for different classification algorithms used in this study.

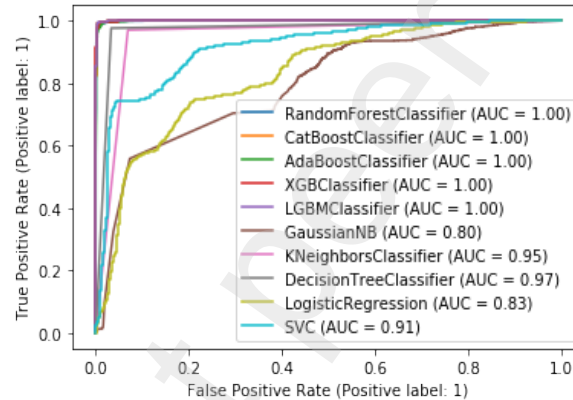


Figure 12: ROC curves for the different algorithms

The aforementioned results reveal that individual classifiers do not provide satisfactory results, but ensemble techniques perform remarkably on Ethereum illegal account detection with 99 % accuracy, which is the highest so far among other research works as per our best knowledge.

Observation 6: Further comparative analysis of ensemble classifiers has been presented in Table 6, Figure 13, 14 and 15, to rank the best ensemble classifier.

Table 6: Performance comparison of developed ensemble techniques

Parameters & Rank	Ensemble Techniques				
	Catboost	Xgboost	LightGbm	Adaboost	Random Forest
Accuracy	99.12	99.18	99.2	98.8	98.9
Rank	3	2	1	5	4
FN Count	10	6	5	11	9
Rank	4	2	1	5	3
Log Loss Value	.31	.28	.25	.41	.4
Rank	3	2	1	5	4
Observed Overall Rank	3	2	1	5	4

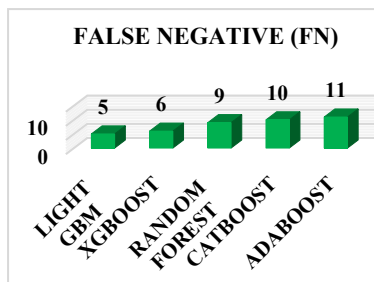


Figure 13

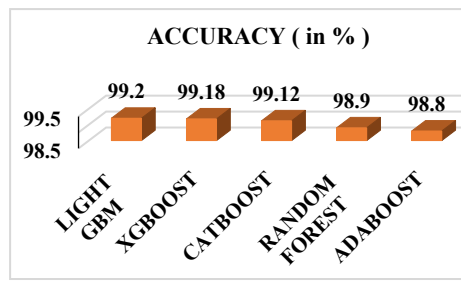


Figure 14

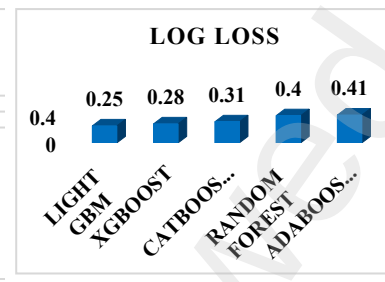


Figure 15

Thus, it can be observed from the aforementioned Table 6 that LightGbm (with accuracy=99.2%) outperforms other ensemble classifiers followed by Xgboost (99.18%), Catboost (99.12%), Random Forests (98.9%) and Adaboost (98.8%).

6. Conclusion and Future Work

In this paper, classification model for detection of illicit accounts over Ethereum networks named as **EnLFADE** has been presented using ensemble learning advances like Catboost, Xgboost, Adaboost, LightGbm, Random forest and some individual classifiers like Naïve Bayes, SVM, KNN, Decision Trees and Logistic Regression. Extracted Kaggle-based Ethereum dataset was initially imbalanced and further it has been balanced using SMOTE in order to achieve an unbiased model. Feature selection played an important role and 20 crucial independent features have been retained out of 33 using correlation analysis. The performances of trained classifiers are evaluated using various evaluation metrics. Results indicate that the ensemble techniques with accuracy approximately 99% provide better outcomes in comparison to individual classifiers. Also, it has been concluded that among the ten aforementioned ML approaches, LightGbm based model with 99.2% accuracy is the best suited for precise detection of the illicit accounts. Extending the outcome of this learning to identify phishing scams over Ethereum network could be an interesting study in this field of research.

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